

LiteCo: Illuminating Workspace Awareness and Social Connectedness with Ambient Display in Home Office

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Figure 1: Showcase the LiteCo system, three colors represent three modes corresponding to different areas in the onsite office: blue for Individual Mode (personal workstations), orange for Group Mode (team meeting areas), and green for Mixed Mode (open workspaces). Remote workers can interact with the system by selecting their current status, getting information about real-time dynamics in the onsite office.

Abstract

With the tendency of hybrid workplace, more people work from home, offering benefits for both individuals and organizations. However, physical separation from colleagues increases social isolation. Home workers often lack information on their colleagues' locations, presence, and intentions—key aspects of Workspace Awareness (WA). WA can contribute to collaborative interactions and have the potential to influence social connectedness. Considering home scenarios, lights within smart homes offer potential as carriers of ambient displays. This study proposed LiteCo, a Hybrid Presence System, designed to promote WA among home workers through lighting design. Twelve participants evaluated LiteCo's impact on WA and social connectedness (SC). Results show that LiteCo can enhance WA by presenting office dynamics to home workers, and lead to higher level of SC. It also helps synchronize work rhythms, fosters emotional connections, and maintains appropriate attention levels while balancing privacy concerns.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing systems and tools.**

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Keywords

Computer-supported collaborative work (CSCW), Hybrid work, Workspace awareness (WA), Social connectedness (SC), Ambient display, Research through Design

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1 Introduction

With the development of communication technologies, the hybrid work model has become a common choice for organizations to address diverse needs [2, 47]. This model offers greater flexibility, improving satisfaction and work-life balance, and benefiting organizations in cost control and productivity [1, 33]. Hybrid workers can choose to work onsite, from home or in other locations [23]. However, remote workers' perceptions of work can decrease due to isolation from coworkers, weakening social connectedness and adversely affecting their well-being [15, 40].

Maintaining social connectedness, a feeling of being concerned for others and belonging to a group or community [21, 43], is essential for the mental health of isolated workers [3, 11]. Awareness systems are used in interpersonal context for supporting social interactions [30], in workplace for facilitating collaborative work [12] and promoting informal communication [35]. This research intends to integrate the insights from previous research to design

awareness systems for home office to support social connectedness among hybrid workers.

Although Gutwin and Greenberg introduced the workspace awareness (WA) framework for collaborative interactions [12], we refer to its awareness elements, specifically where they are and what they are doing. Acquiring workspace awareness is easy in-office but challenging remotely. While media like video conferencing and instant messaging can facilitate remote workers' perception of social presence, some invisible connections are still missing [18].

For remote workers, the home often serves as their primary workspace. With advances in Internet of Things (IoT) technology, smart home devices now interconnect for automation and remote control [19, 26, 27]. The rise of hybrid work has led smart home technology to play an essential role in home office environments [29]. Connecting hybrid workplaces through technologies can help reduce the feeling of isolation and enhance the rhythm of work [28]. Ambient lighting, previously used to foster social connectedness among friends and families, can also be employed to convey workplace information [39, 41]. Thus, our design case focused on using ambient lighting to communicate colleagues' activities to hybrid workers.

This paper adopts a Research through Design (RtD) approach [46] to investigate how ambient lighting displays can promote workspace awareness and foster social connectedness in the home office. By integrating smart home technologies into hybrid work scenarios, our design aims to create a social experience for hybrid workers.

2 Related work

2.1 Challenges of Social Connection in Hybrid Work

Accenture reports 83 percent of employees globally prefer hybrid work for its flexibility [1, 2]. Work-from-home (WFH) remains the most popular teleworking method, promoting efficiency and work-life balance but reducing social interactions [4, 23, 33, 40, 47]. According to Self-Determination Theory (SDT), fostering social connections addresses the fundamental psychological need for relatedness, which enhances job performance and well-being [14, 31]. Lack of in-person interactions can lead to loneliness and reduced belonging [3, 40]. Corporations such as GitLab recognize the importance of workplace connection, they developed a virtual coffee break to facilitate informal communication for establishing identity and presence in virtual co-working environments [15, 22, 32]. Connections and interactions among people, known as social connectivity, are measured through social connectedness—the feeling of closeness or connection with others [3, 7, 10, 21, 34]. It encompasses dimensions including relationship saliency, closeness, contact quality, knowing each other's experiences and shared understanding [41]. While hybrid work offers benefits, it reduces social connections, and efforts to improve remote connectivity are ongoing.

2.2 Workspace awareness and social context display

Strong social connectedness and organizational support improving remote workers' well-being [11]. Awareness information, including

feelings, experiences, emotions, and activities of other people, is critical to improving the sense of belonging and connectedness [36]. In work scenarios, academics have proposed the concept of workspace awareness (WA), which involves the understanding of others' work environments, their current tasks, and upcoming actions. Such information improves collaboration through task coordination and anticipation [12]. WA has proven effective in distributed teams, as seen in the DIVA virtual office, which boosts teamwork by real-time activity updates [24]. In addition, social translucence proposes making activities visible while protecting privacy, enabling informed decisions and fostering accountability [17]. Therefore, the selection of information could be linked to make presence and activity visible or perceivable, to provide social cues that allow people to understand and make choices, subsequently structuring the activities [12, 28]. Social context display serve as tools to convey social and activity cues for supporting decision making. For example, the NOMATIC system visualizes user states in shared spaces, allowing interaction via gaze while protecting privacy [16]. AmbiTeam maps team activity levels to reflect work intensity and collaboration dynamics [42].

2.3 Peripheral interaction with lighting

Awareness displays often operate in a peripheral area, focusing on lightweight and informal communication [16, 24]. As ubiquitous computing grows, concerns about information overload have emerged. Hence, scholars propose "calm technology", which emphasizes peripheral attention—a state allowing individuals to sense surrounding while focusing on primary tasks [6, 44]. Ambient displays utilize peripheral interaction to present non-critical information, delivering meaningful information in a cognitively friendly way [8, 37]. Light can serve as helpful medium for providing feedback on the state of the systems in the environment, becoming a possibility for peripheral interactions. Present technology allows LEDs to be integrated into digital controls, adjusting the color or pattern of light can provide subtle feedback and sufficient information [5, 45]. Light influences people both physically and psychologically, with creating atmosphere, shaping emotions, and regulating their social activities [20, 38]. In addition, light can convey a variety of information with different semantics in diverse contexts with wide-ranging applications, such as supporting remote emotional interactions for the elderly, visualizing home energy and encouraging conservation practices [13, 27, 45].

2.4 Opportunities for IoT smart lighting with hybrid work

With the support of computer-supported collaborative work (CSCW), home office as a distributed work scenario, is one of the most common choices for hybrid work [4, 23]. In the home scenario, the smart home is prevalent as Internet of Things (IoT) technology emerges, but few IoT system interaction aims to raise awareness [19, 27]. Scholars propose that linking remote workplaces through environments bolsters a sense of community and co-presence, minimizes the feeling of isolated work, and brings a sense of pace to the workday [28]. However, the exploration of ambient media, such as lighting, to help remote workers perceive the workplace dynamics and enhance social connectedness is limited.

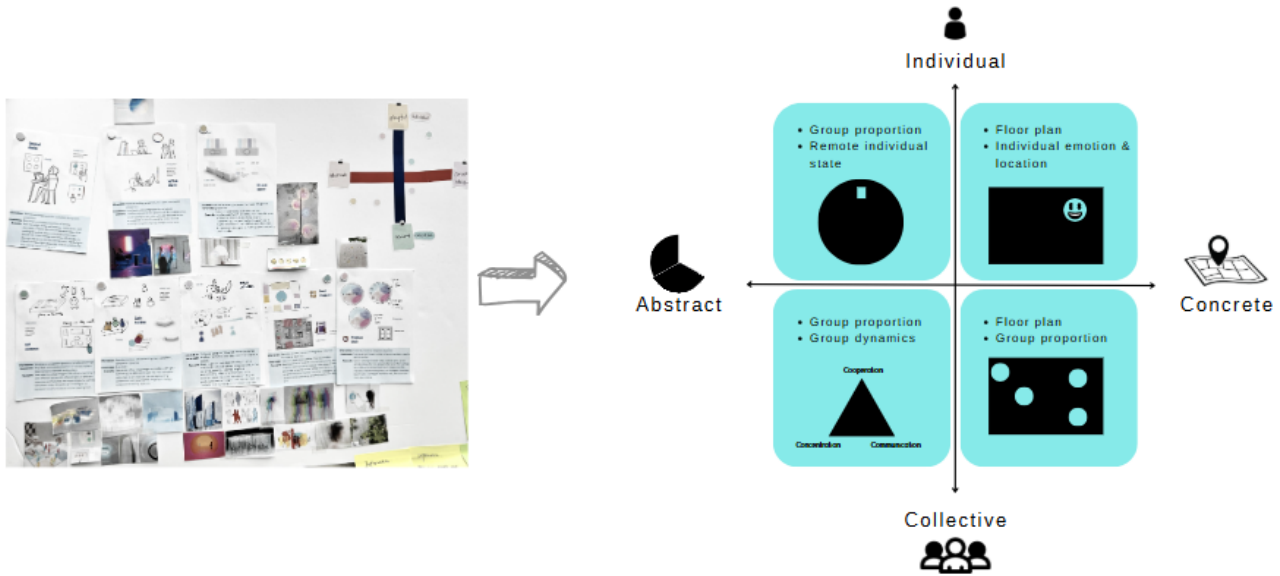


Figure 2: The framework organizes design concepts along two axes: Individual vs. Collective, distinguishing whether information relates to a single person or a group, and Abstract vs. Concrete, reflecting the level of detail or ambiguity in the representation. It was used to map and refine ideas, balancing privacy with awareness and guiding the development of LiteCo’s ambient display system.

3 DESIGN AND IMPLEMENTATION

This section outlines the design and implementation process of LiteCo. Beginning with brainstorming, ideas were organized in a conceptual framework, and then we developed concepts into prototype.

3.1 Design Rationale

Building on prior research, this research explores the practice of lighting design in bridging the smart home and hybrid work domains. We intend to create a supportive and inclusive hybrid work environment that promotes home workers’ awareness of onsite workers’ social presence. Light, as an environmental stimulant, can create atmosphere, evoke emotions and provide immersive experiences. Thus, we chose to use lighting as the medium for designing ambient systems to enable home office workers to intuitively perceive office dynamics. [20, 37, 38].

3.2 Ideation and Conceptual Framework Development

We started with brainstorming design ideas, exploring information displays and interactions. By examining what information can be conveyed, we characterized concepts in two dimensions: Individual vs. Collective, describing the information linked to individual or group; Abstract vs. Concrete, emphasizing the ambiguity level of information. Figure 2 illustrated that we used this framework to organize our design ideas and guide further exploration.

3.3 Low-Fidelity Prototyping and Concept Refinement

We developed four concepts with low-fidelity prototypes by mapping the two dimensions in the framework (as shown in Figure 3). Concept 1 involved a lighting product with a floor plan where users could input their state by placing emojis in specific locations, creating a colorful display. Concept 2 assumed a desk device to show the busyness of different areas with virtual representations. Concept 3 looked like a clock, lighting up areas to represent the crowdedness and allowing users to check if others were remote. Concept 4 used three circles to display areas, enabling home workers to check individual states for remote coworkers. By reviewing these concepts with external researchers, we identified key issues:

- **Clarity for Relationships:** People wanted to understand the relationship and transitions among the three types of information.
- **Privacy:** Revealing specific individuals’ information can be intrusive.
- **Generality:** Designs adaptable to various company needs are more versatile than those limited to specific layouts.

3.4 Final Design

Based on the above insights, we finalized a concept centered on visualizing group traffic flow in the office. Our design defined three interrelated and switchable social states: group, individual, and in-between. These states were derived by observing that different office areas serve distinct functions, as people’s locations naturally conveying information about their activities. Using location data allowed us to balance information delivery and privacy, as the

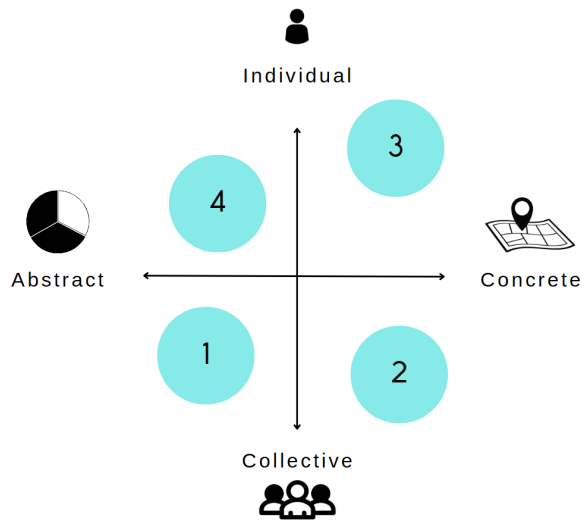


Figure 3: The low-fi prototypes mapped four concepts onto the framework’s axes: Individual vs. Collective and Abstract vs. Concrete, with each concept positioned in a distinct quadrant.

collective information collected from the environment abstracted individual details. Additionally, dividing information by general zones rather than relying on a specific floor plan makes the design more adaptable to various organizations and workplaces.

To represent these states and their transformations, we used a triangle. Early prototypes with laser-cut triangles and multi-color lighting highlighted challenges like information overload (see Figure 4). After further discussion, we simplified the design to a triangle for showing dynamic relationships and a circle for aesthetic harmony. This approach effectively conveys the fluidity between social states while maintaining visual simplicity, ensuring adaptability and fostering better workspace awareness.



Figure 4: The early prototype, created through laser cutting, served as a foundational model for the final LiteCo system.

3.4.1 System design. LiteCo is a hybrid presence system designed to support workspace awareness and enhance social connectedness for home workers. The system collects traffic density and flow information from the office environment, allowing home workers

to view area crowdedness and overall dynamics through interaction. Three modes were identified based on office areas (see Figure 5):

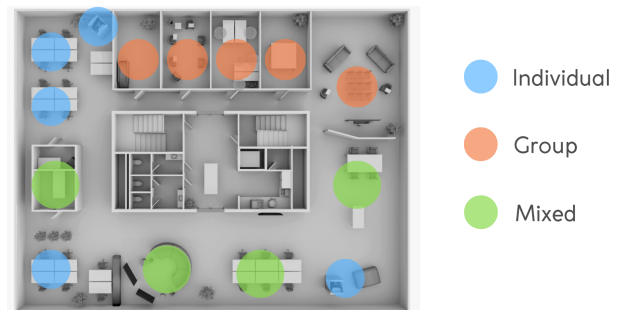


Figure 5: Example of office zoning, with a floor plan divided into three areas: individual workspaces (blue), group meeting zones (orange), and mixed collaborative spaces (green).

- **Individual:** Solo areas such as individual workstations or relaxation zones.
- **Group:** Meeting areas where teams typically gather.
- **Mixed:** Open workspaces where people may work alone or engage in informal communication.

Lighting colors were chosen based on their psychological effects. Research revealed that cold colors occupy very little attention and lead to feelings of calmness; whereas warm colors were appealing and could be used to evoke energy [9, 38]. As a result, we decided that Individual was in blue tones for personal focus, Group in orange tones for evoking a sense of socialization, and Mixed (In-between) in green tones to give a sense of openness.

3.4.2 Technical Implementation. Final realization mainly focuses on light processing, we connected two light strips for LiteCo. The first was inside the triangle, with a flowing light effect to show

the traffic dynamics; the second one was within the circle, with a gradient effect to show the change of the crowds' denseness. The ESP32 board, with integrated WiFi, was selected for remote control. An external power supply was used to support the light strips. Arduino IDE was used to develop local web pages to control the ESP32 board, offering five light effects to demonstrate functionality.

4 Method

To investigate how the LiteCo system influences workspace awareness and social connectedness of remote employees, our study intended to gather insights across these dimensions:

- How participants interact with LiteCo system, in terms of what they perceive, how they interpret the information, and with the intended further action.
- How the information from the system supports participants in obtaining insights about the office, and the consequential impacts on social connectedness.
- Participants' attitudes toward the hybrid presence system and their experience with the design.

4.1 Participants

We recruited 12 master students (N=12), with 5 males and 7 females, aged 22 to 28 years. All had internship or work experiences, as well as hybrid studying experiences. The experiment was approved by the university ethical review board. All participants read and signed a consent form before starting the experiment.

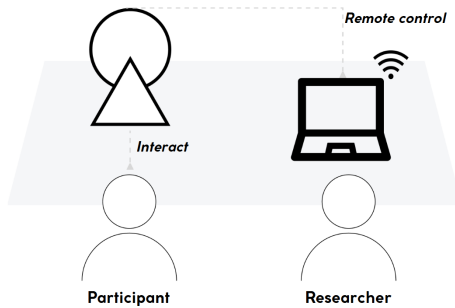


Figure 6: Test set-up

4.2 Experimental Settings and Measures

We conducted a qualitative experiment in a distraction-free classroom, simulating a home office. The controlled environment standardizes variables such as lighting and participants' interactions to ensure consistency. We employed the Wizard-of-Oz technique to simulate the system functionality. As shown in Figure 6, the researcher operated the system remotely during participants' interactions.

Assessing social connectedness requires longer interactions, making it hard to measure in short-term experiments. This study examines workspace awareness, reflecting perceptions of hybrid environments and supporting social connectedness. The experiment consisted of two parts (Figure 7):

Experiment. Participants were briefed, introduced to a virtual worker (Ryan), and completed tasks interacting with LiteCo.

Observations and verbal feedback were recorded to learn about how they were aware of the workplace. The structure for the observation sheet was inspired by the workspace awareness framework, which is a loop among "environment-workspace awareness-action/exploration" [12]. Afterward, we conducted a semi-structured interview covering workspace awareness, its impact on social connectedness, information clarity, interaction and attention, and overall experiences.

Focus Group. Participants formed three teams to discuss their favorite features, desired changes or additions, and the impact of information about onsite office. The goal was to enhance understanding and gather diverse perspectives on workspace awareness and social connectedness. Participants also explored what they wanted to know about the office and what they wished others to know about themselves, complementing each other's viewpoints.

5 Finding

We used thematic analysis [25] to code and categorize the themes from the interviews, observations, and focus group data.

5.1 Synchronized Work and Efficiency Improvement

Participants could interpret the ambient lighting changes and understand the information under the guidance of instruction. When knowing about the lighting flows represent traffic flows and the shades of color represent different crowded level color shades signify crowding levels, participants could imagine and explain the dynamics of onsite office. Such insights about office could help people to better speculate about others' states, actions and activities, resulting in enhanced workspace awareness (e.g., "This system makes me feel like I'm staying with the crowd", P5; "I feel more connected to others and like I'm in the onsite office", P12). On the other hand, lighting indication supports home workers obtain rich implicit information by guessing, such as the locations where coworkers gather and the overall office atmosphere. The system also fosters emotional connection by helping remote workers view their home as part of a cohesive workplace. This emotional engagement strengthened hybrid workspace awareness for remote workers.

5.2 Enhanced Workspace Awareness through Ambient Display

We found participants appreciated the synchronized states with co-workers. When users selected a state, the lighting ambience adopted a unified hue. Participants reported that such a tone fostered productivity by creating a focused atmosphere. Additionally, testers indicated that they might adjust their pace to align with onsite colleagues (e.g., "My work schedule will be affected slightly, it like a time reminder that reminds me of what to do and at what time, and will make me adjust my work status with the colleague in the office", P1). Consistency benefited participants psychologically, as they noticed being in the majority or minority within the same state, thereby coordinating their work. People believed that such synchronization had positive effects on their productivity and collaboration. The home office was no longer isolated, and mutual knowledge of the Workspace made it possible for employees to initiate more informal interactions.

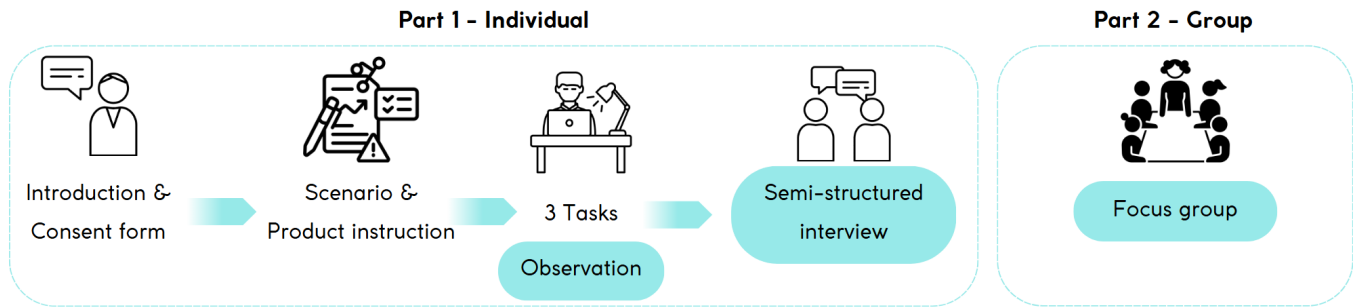


Figure 7: Visual representation of the study procedure

5.3 Emotional Connection and Team Dynamics with Lighting Dynamics

People believed that the system could increase social interactions and bring people close to each other emotionally. The system could provide availability information for home workers to participate in small talk. A possible scenario is that a home employee observes many people flowing into the in-between area might sense a relaxed office atmosphere and initiate an online conversation with a familiar colleague (e.g., “*The system promotes my interactions with others: if there are a lot of people at in-between state, I might guess that people may have group activity, and ask someone I am familiar with to learn about what is going on.*”, P7). The sense of connectedness among coworkers is mutual. On one side, remote people are informed of office dynamics through light indicators. Visualizing workplace dynamic helps users sense the team atmosphere and enhances emotional connection.

5.4 Appropriate Attention Occupation

Most participants claimed that the system would not distract them but rather give moderate reminders to help keep the working pace (e.g., “*It is not distracting to work and instead it reminds me to keep same pace with the office*”, P7). Their attention shifts to the system when required to understand the dynamics. In line with the peripheral interactions principles, this system allows people to perform their primary tasks while maintaining sensitivity to the dynamics of onsite office, seamlessly shifting attention between central and focus.

5.5 Information Ambiguity and Privacy Concerns

Some participants expressed concerns about privacy, noting that many people choose to falsify states (e.g., “*I might fake information because I don’t want people to potentially know my exact status. Even though it is highly anonymized*”, P5). Hence, the system needs to find a better trade-off between privacy protection and information specificity. While abstract and ambiguous information provided at present is insufficient to support more socialization activities; on the other hand, too much exposure of concrete information may lead to privacy violations. Overall, people find the current level of abstraction acceptable, as home employees access collective office information.

6 Discussion and Conclusion

We propose LiteCo, a hybrid presence system utilizing connected lighting to support home workers to better understand workplace dynamics. This system challenges social isolation and has positive effects in enhancing work synchronization and social connectedness.

However, there are limitations to design and experiment. The overlapping circular and triangular design is innovative but sometimes causes confusion in indicating the traffic density and flow. In addition, the color scheme could be more intuitive to reduce the learning effort. Short-term trials limit the exploration of social connectedness, which requires over the long term, and assessment in a controlled campus environment lacks real-world validation.

Future improvements include refining form and visual design. For example, exploring more geometric possibilities and customizable color schemes for increased usability and aesthetics. The product’s extensibility also should be considered. One possibility is to add or change elements that adapt to other remote workspaces beyond the home office. For instance, adding a wearable device or small desktop accessory which can better collect inputs and cover more scenarios outside of the onsite and home office. Another is to consider accessing some office software or calendars, so that the product can cover more utilization situations.

Overall, LiteCo contributes to multiple aspects. It fosters a supportive and connected working environment for distributed teams, reducing loneliness and social isolation of telecommuters. Integrating knowledge of peripheral interactions and ambient display into design, LiteCo provides subtle updates on workplace dynamics and creates aesthetics and emotional value. It bridges the domains of smart home technology and hybrid work to improve users’ psychological well-being and work experiences through non-intrusive design, creating a new design perspective. The Research through Design approach generated practical insights, including a design framework balancing abstract and concrete information and individual versus collective awareness, offering guidance for future explorations in workspace awareness and social connectedness. Finally, LiteCo demonstrates the potential of IoT systems to enhance hybrid work environments, extending their application beyond traditional smart home scenarios.

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